👩‍🏫 **Teacher Copy — includes the Answer Key & Teacher Guide. Do not distribute to students.**

UNIT 2 · STANDARD 6.DS.5

Represent and Describe Data in a Histogram

Lesson 2-2 · Teacher Copy

Name: _____ **Date:** _____ **Class:** _____

My Notes

Level 3 Enrichment

🎯 **TODAY'S OBJECTIVES**

Content Objective: I can make and read a histogram to display data in intervals.

Language Objective: I can explain my histogram using the words histogram, frequency, interval, and distribution.

📖 See **Learn It** for the explanation and worked examples, and the **Vocab** tab for the words. These are your notes to fill in and keep.

✏️ **Fill in each blank as we go. Use the Word Bank to help you.**

📚 **WORD BANK — FILL EACH BLANK WITH THE BEST WORD**

Histogram · **Frequency** · **Interval** · **Distribution** · **Variability** · **Frequency table** · **Shape of the data** · **Peak**

Level 3: try the blanks from memory first, then check the bank.

1 A bar graph that shows how often data falls into equal intervals is a _____.

2 How many data values fall in an interval is the _____.

3. An equal-sized range of values used to group data is an _____.

4. The overall pattern of how data is spread out is the _____.

5. How much the data values differ is the _____.

6. A table that lists each interval and its count is a _____.

7. The pattern the bars make across a display is the _____.

8. The tallest bar in a display is the _____.

Write About the Math

Does the histogram support the coach's claim? What does the shape of the data tell you?
¿El histograma apoya lo que dice el entrenador? ¿Qué te dice la forma de los datos?

Say your answer to a partner first. Then write it, using at least two of these words:

- ☐ **Histogram** *Histograma*
- ☐ **Frequency** *Frecuencia*
- ☐ **Interval** *Intervalo*
- ☐ **Distribution** *Distribución*
- ☐ **Variability** *Variabilidad*

Model: *A histogram groups data into equal intervals so large data sets become readable.* — Students explain grouping into intervals makes 30 values readable and that each bar's height is the frequency for that interval.

Start

Complete the frames. Use the word bank.

Completa las oraciones. Usa el banco de palabras.

- **The histogram peaks at the ___ interval with ___ players. The shape is approximately ___. This ___ the coach's claim because ___.**
- **My answer is ___ because ___.**
  Mi respuesta es ___ porque ___.

Build

Write two connected sentences. Use because, so, or but.

Escribe dos oraciones conectadas. Usa porque, entonces o pero.

- **My claim is ___.**
  Mi afirmación es ___.
- **My evidence is ___, so ___.**
  Mi evidencia es ___, entonces ___.

Explain

Write a claim, give math evidence, and explain your reasoning.

Escribe una afirmación, da evidencia matemática y explica tu razonamiento.

_____

_____

_____

_____

_____

- [ ] I answered the question.
- [ ] I used math evidence: numbers, an equation, or a model.
- [ ] I used at least two math words.

Writing Guide — Teacher Copy

- The answer to the math question is correct.
- The evidence is real lesson mathematics (numbers, equation, or model), not restated claim.
- The support level changed the language scaffolding, never the mathematical expectation.

Answer Key & Teacher Guide

1. **Notes 1:** Histogram
2. **Notes 2:** Frequency
3. **Notes 3:** Interval
4. **Notes 4:** Distribution
5. **Notes 5:** Variability
6. **Notes 6:** Frequency table
7. **Notes 7:** Shape of the data

8. **Notes 8:** Peak
9. **Watch for this mistake:** *A common mistake in Display Data: Histograms is counting a boundary value in two intervals at once — for example, putting the value 10 in both the '0–9' bin and the '10–19' bin. Each data value belongs to exactly ONE interval. Before you submit, add up all your frequencies and check that the total matches the number of data values you started with.*
10. **Try It 1:** B. 10–19 with 9 players — *The 10–19 interval has the greatest frequency (9 players), so its bar is the tallest.*
11. **Try It 2:** D. 16 — *Add every bar's frequency: 2 + 8 + 5 + 1 = 16 players total.*